Microprocessor & Computer Architecture (µpCA)

UE24CS251B

Session - 3

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ARM program structure and Introduction to Arm Simulator

Address	Instruction & Data

.text

Address of Instruction 1
ARM Instruction_1
Address of Instruction 2
ARM Instruction_2
……………………
……………………
……………………
……………………
Address of Instruction n
ARM Instruction_n

.data

Address of Data1
Declaration of variable 1
Address of Data2
Declaration of variable 2

Note: Blue color depict the code written by the programmer
Orange color depict the address assigned during execution

General Format of Instruction

MNEMONIC{condition}{S} {Rd}, Operand1, Operand2

MNEMONIC - Short name of the instruction. ***Eg: ADD, SUB...***

{condition} - Condition that is needed to be met in order for the instruction to be executed ***EG: EQ, MI, GT, LT, LE, AL, NE***

{S} - An optional suffix. If S is specified, the condition flags are updated on the result of the operation . ***Eg: To set N, O, C, V of CPSR***

{Rd} - Register (destination) for storing the result of the instruction

Operand1 - First operand. Either a register or an immediate value

Operand2 - Second (flexible) operand. Can be an immediate value (number) or a register with an optional shift

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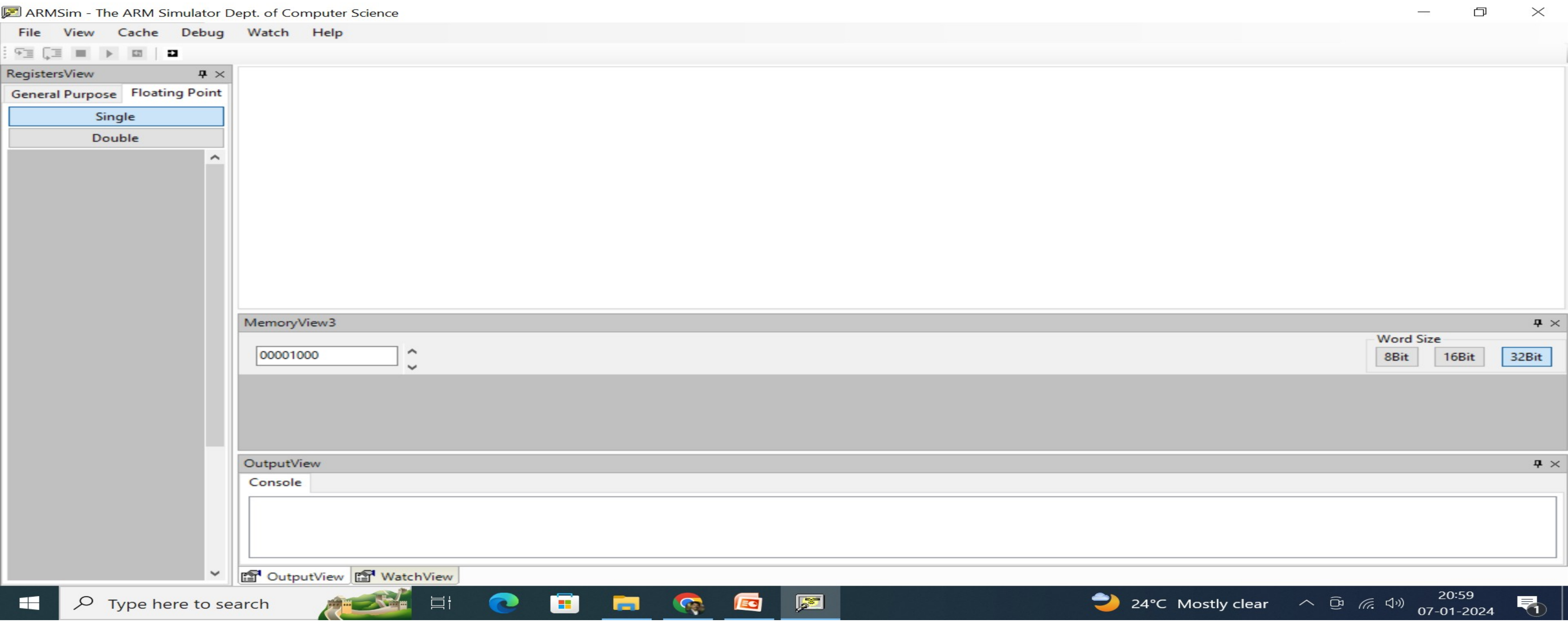
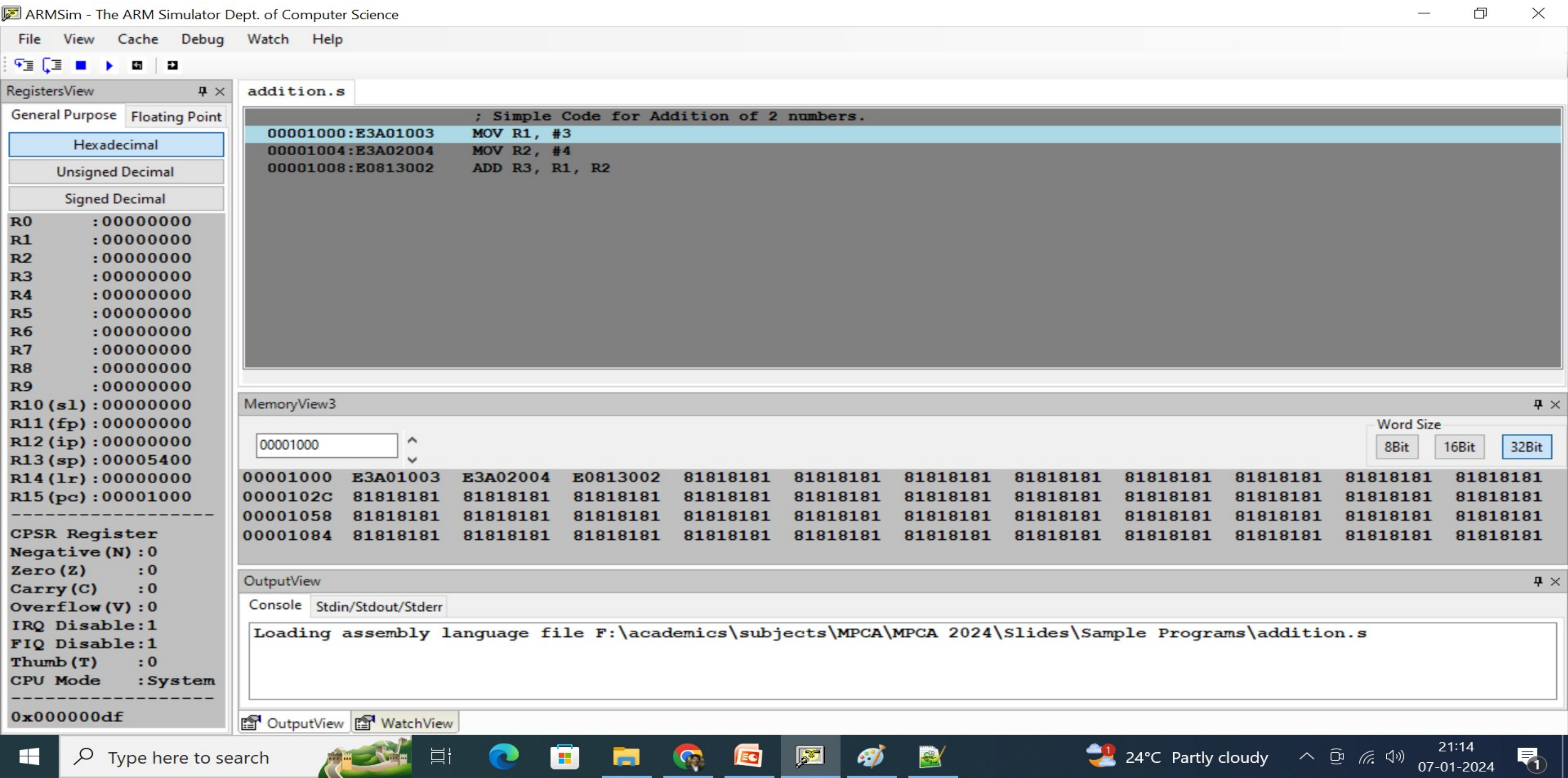
ARMSIMULTOR



NOTE: Recommended to use and demo on ARM simulator directly.

Loading Sample Program on ARMSIMULATOR



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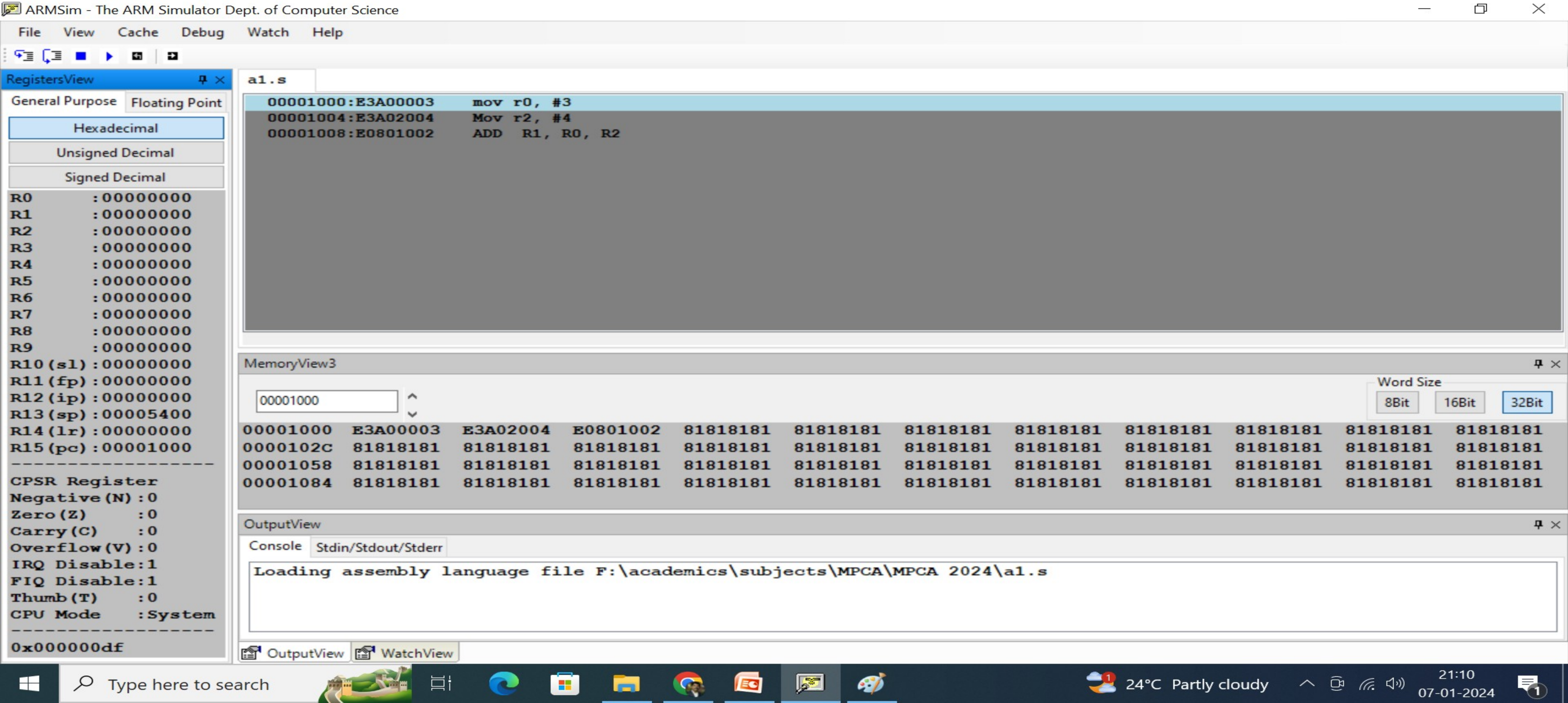
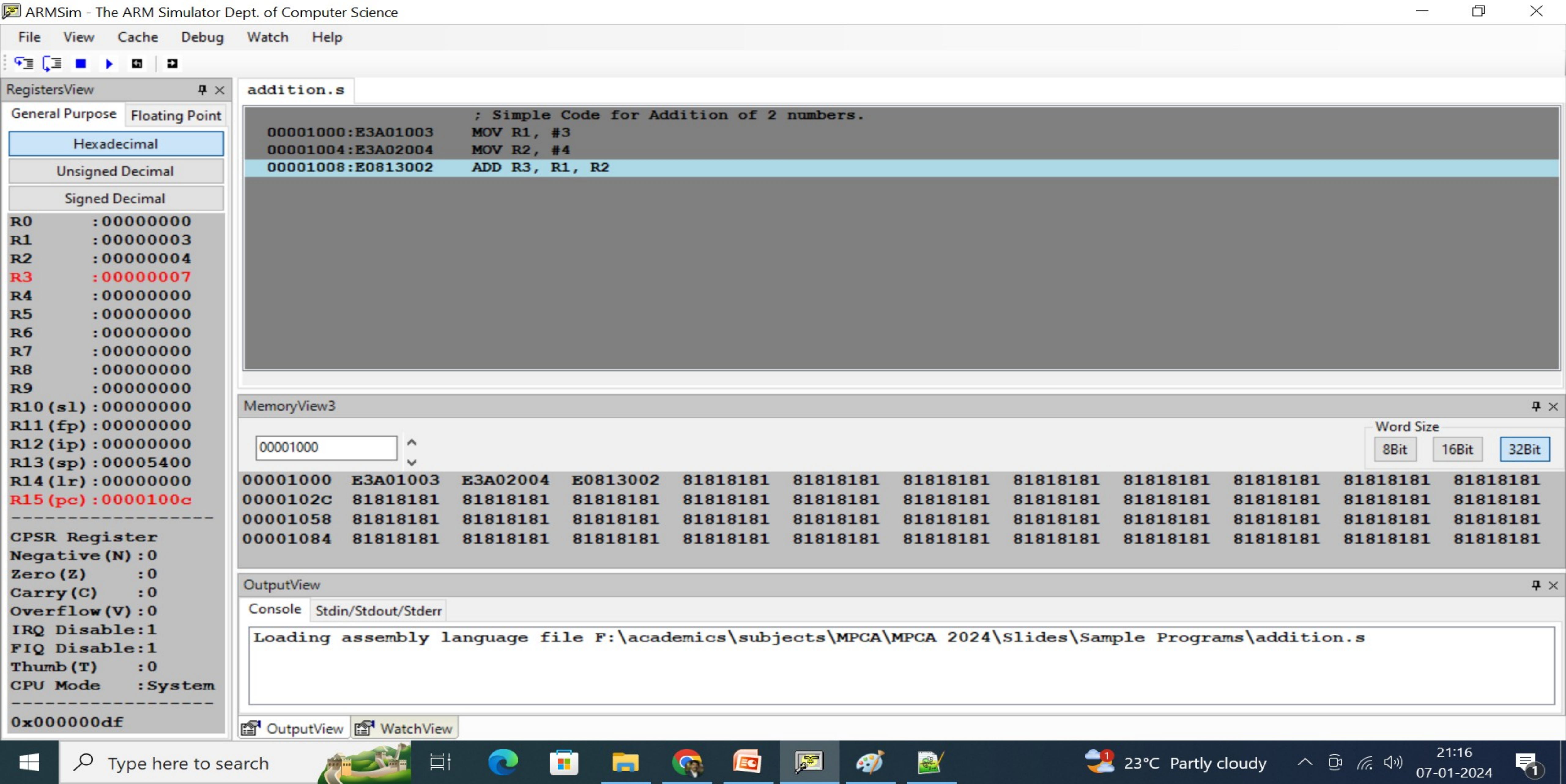
Register View (ARMSIM Simulator)



General Format of Instruction Example

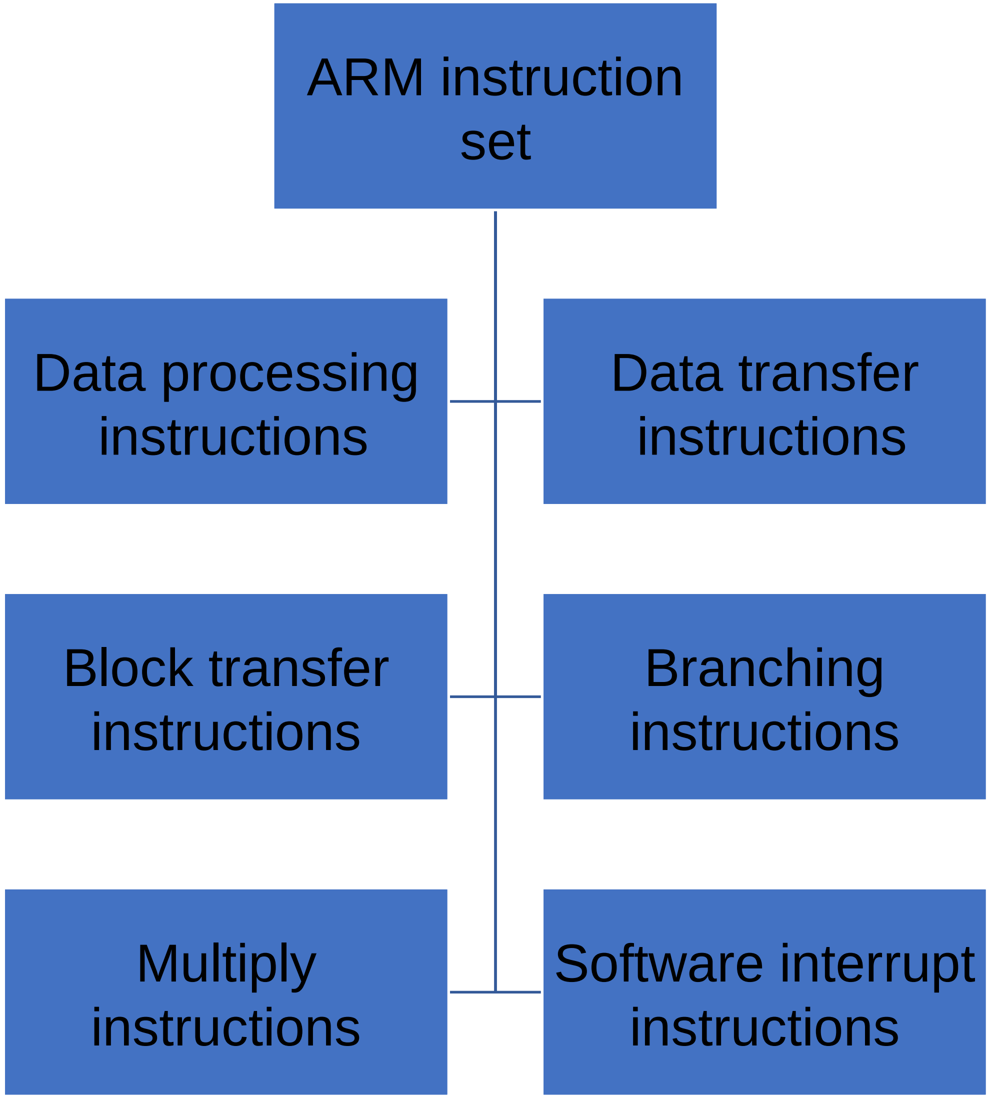


Samples of following types of Numerical Data on Simulator

Demonstration of the following data is shown in the simulator.

- Decimal Numbers
- Hexadecimal Numbers
- Octal Numbers
- Usage of Bytes, Decimal and Hexadecimal data in the code and in the Simulator
- Read and observe the contents of the following in the simulator.
  - **Memory**
  - **Stack**
  - **Registers**
  - **Flag Register**
  - **Console**

ARM Instruction Set



THANK YOU

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